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INFORMATION TECHNOLOGY**

**BACHELOR OF COMPUTER SCIENCE (HONS)
BACHELOR OF COMPUTER SCIENCE (HONS) IN CYBERSECURITY**

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Student's Signature: 

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PART 1: CRYPTOGRAPHIC SYSTEM REPORT

1.1 Introduction

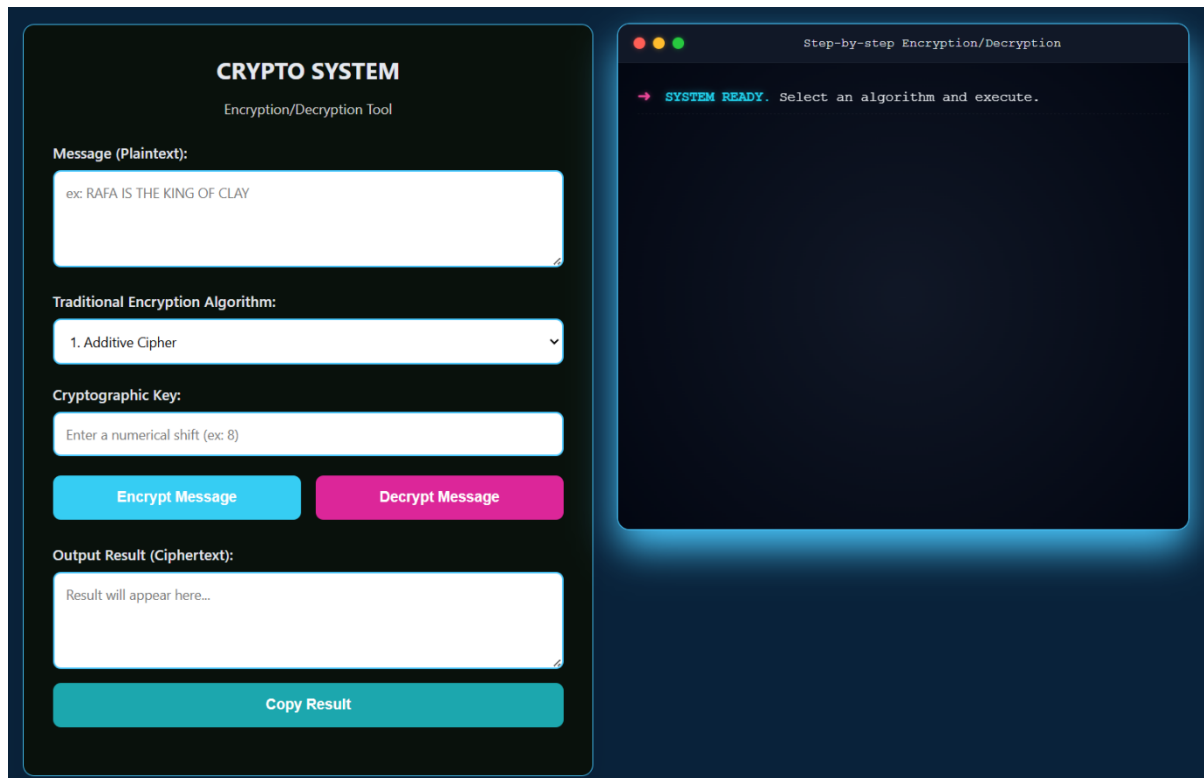
In the contemporary cybersecurity landscape, cryptographic techniques remain the primary means of enforcing data confidentiality, integrity, and authenticity. This project implements a browser-based cryptographic system that utilises traditional, symmetric encryption algorithms. By offering an interactive environment where users can encrypt and decrypt messages using multiple classical ciphers, the system illustrates the practical foundations upon which modern cryptography is built. The tool not only demonstrates algorithmic substitution and transposition but also reinforces the importance of proper key management and user-friendly security interfaces.

1.2 System Design and User Interface

The system was developed using **HTML5, CSS3, and JavaScript** – ensuring zero external dependencies and allowing instant execution in any modern web browser. During development, ChatGPT (GPT-5.5) was utilized to assist with structural CSS/HTML referencing and UI animation optimization (OpenAI, 2026). The graphical user interface (GUI) follows a responsive, two-column layout:

- **Left Control Panel**
Contains the plaintext input area, a dropdown for cipher selection (Additive, Simple, Vigenère, Rail Fence), a dynamic key input, and action buttons (Encrypt, Decrypt, Copy Result). Real-time validation provides immediate feedback on invalid keys.
- **Right Visual Console**
A stylised terminal window (MacOS-inspired) that **animates every algorithmic step** in real time. Each character transformation is displayed with colour-coded markup, enabling the user to trace exactly how the ciphertext was derived or reversed.

The interface adheres to fundamental usability principles: clear labelling, logical grouping, and visual feedback. All buttons, menus, and links are fully functional, and the dark theme reduces eye strain during prolonged use.



[Figure 1: Full System Interface – showing the two-panel layout with a sample encryption]

1.3 Algorithms Used in the Cryptographic System

Four classical encryption techniques were implemented. Each algorithm handles only alphabetic characters (A–Z); spaces, digits, and punctuation are preserved unchanged, reflecting real-world example practices.

1.3.1 Additive (Caesar) Cipher

A mono-alphabetic substitution cipher that shifts each letter by a constant k modulo 26.

- **Encryption:** $C = (P + k) \bmod 26$
- **Decryption:** $P = (C - k + 26) \bmod 26$

The system validates that the key is numeric and automatically wraps shifts exceeding 26.

1.3.2 Simple (Keyword) Cipher

A substitution cipher that builds a custom alphabet from a user-provided keyword. After removing duplicate letters from the keyword, the remaining standard alphabet letters are appended, forming a 26-character permutation.

- **Encryption:** Maps standard alphabet positions to the cipher alphabet.
- **Decryption:** Performs the inverse mapping.

The key must consist solely of letters; the interface rejects non-alphabetic input.

1.3.3 Vigenère Cipher

A poly-alphabetic cipher that uses a repeating keyword. Each plaintext letter is shifted by the numeric value of the corresponding keyword letter.

- **Encryption:** $C_i = (P_i + K_{i \bmod m}) \bmod 26$
- **Decryption:** $P_i = (C_i - K_{i \bmod m} + 26) \bmod 26$

The key index advances only when a letter is processed, preserving the integrity of non-alphabetic characters in the ciphertext.

1.3.4 Rail Fence Cipher

A transposition cipher that writes the plaintext in a zig-zag pattern across a specified number of “rails” and then reads off each rail sequentially.

- **Encryption:** Builds a 2D fence matrix, places characters row-by-row according to the zig-zag motion, then concatenates rows.
- **Decryption:** Marks the positions in the fence, fills them with the ciphertext characters in row order, and reconstructs the original message by following the zig-zag path.

The key is an integer ≥ 2 , validated by the system.

All algorithms are documented with inline code comments and are implemented in a concise, modular fashion, demonstrating **minimal yet highly readable code design**.

1.4 Demonstration Examples

To validate system correctness, multiple encryption and decryption examples were executed. Below are **three encryption** and **three decryption** samples, each illustrating distinct ciphers and key types.

1.4.1 Encryption Demonstrations

Sample	Traditional Cipher Method	Plaintext (Messages)	Key	Result (Ciphertext)	Step-by-Step Console Trace
1	Additive	KING RAFA	3	NLQJ UDID	K→N, I→L, ... (Space preserved)
2	Vigenère	HELLO WORLD	ABC	HFNL YOSND	H+0→H, E+1→F, L+2→N, L+0→L, ...
3	Rail Fence	SEGI UNIVERSITY	2	SG NVR IEEIUIEST	Row0: S,G, ,N,V,R,I,E Row1: E,I,U,I,E,S,T

```

Step-by-step Encryption/Decryption

→ INITIALIZING ADDITIVE ALGORITHM...
→ ADDITIVE Mode: Encrypt | Shift: 3
→ Letter 'K' (10) + 3 = 13 → 'N'
→ Letter 'I' (8) + 3 = 11 → 'L'
→ Letter 'N' (13) + 3 = 16 → 'Q'
→ Letter 'G' (6) + 3 = 9 → 'J'
→ Preserved non-letter: ' ' (unchanged)
→ Letter 'R' (17) + 3 = 20 → 'U'
→ Letter 'A' (0) + 3 = 3 → 'D'
→ Letter 'F' (5) + 3 = 8 → 'I'
→ Letter 'A' (0) + 3 = 3 → 'D'

→ Encryption / Decryption process complete.
  
```

[Figure 2: Console output for Example 1 – Additive encryption step by step]

1.4.2 Decryption Demonstrations

#	Cipher	Ciphertext	Key	Result (Plaintext)	Step-by-Step Console Trace (Selected)
1	Simple	RCKKFQ	BLUEC AT	TENNIS	Inverse matrix mapping (T→R, E→C, ...)
2	Additive	KHOOR	3	HELLO	K(10)-3=H, H(7)-3=E, ...
3	Vigenère	CMPY VCXE	SEC	CYBER SECURITY	U-S→C, A-E→Y, F-C→B, ...

```

Step-by-step Encryption/Decryption

→ INITIALIZING SIMPLE ALGORITHM...

→ SIMPLE Mode: Encrypt | Key: BLUECAT

→ Base Matrix: ABCDEFGHIJKLMNOPQRSTUVWXYZ

→ Cipher Matrix: BLUECATDFGHIJKLMNOPQRSVWXYZ

→ Encode: 'T' → 'R'

→ Encode: 'E' → 'C'

→ Encode: 'N' → 'K'

→ Encode: 'N' → 'K'

→ Encode: 'I' → 'F'

→ Encode: 'S' → 'Q'

→ Encryption / Decryption process complete.

```

[Figure 2: Console output for Example 1 – Simple decryption step by step]

1.5 Conclusion (Part 1)

The implemented cryptographic system successfully meets the objectives of Part 1. It combines an attractive, fully functional interface with a rich algorithmic backend that supports four traditional encryption ciphers. The step-by-step animated console transforms abstract mathematical processes into a tangible learning tool, clearly illustrating how confidentiality can be enforced through encryption. The project demonstrates a solid understanding of fundamental cybersecurity concepts – notably substitution and transposition – and provides a robust platform for further expansion into modern cryptographic standards.

PART 2: CYBERSECURITY POLICIES AND PROCEDURES

2.1 Introduction

Technology alone cannot guarantee information security; human behaviour is equally critical. This section presents a set of **five security policies**, each targeting a core pillar of cybersecurity: **Confidentiality, Integrity, Availability, Authentication, and Authorization** (CIA + AA).

For each policy, concrete procedures, monitoring measures, and consequences of non-compliance are defined. These documents are designed to be distributed to all company personnel, ensuring a shared understanding of security responsibilities and fostering a culture of proactive defence (Whitman & Mattord, 2021).

2.2 Policy 1 – Confidentiality: Data Classification and Handling

Purpose:

To protect sensitive company information from unauthorised access and disclosure.

Scope:

All employees, contractors, and third-party partners who create, store, or transmit company data.

Policy Statements:

1. All data shall be classified as **Public, Internal, or Highly Confidential** based on sensitivity.
2. Highly Confidential data (PII, trade secrets, financial records) must be encrypted at rest (AES-256) and in transit (TLS 1.3) (Stallings, 2011).
3. Physical documents containing Highly Confidential information must be locked in secure cabinets when not in use and shredded upon disposal.
4. Disclosure of Highly Confidential data to external parties requires a signed Non-Disclosure Agreement (NDA) and prior approval from the Data Protection Officer.

Procedures:

- **Data Labelling:** Department heads will label all new assets within one week of creation.
- **Encryption Enforcement:** The IT department will deploy endpoint encryption software and monitor compliance monthly.
- **Clear Desk/Clear Screen:** All unattended workstations must be locked; spot checks will be conducted weekly.
- **NDA Workflow:** The legal team will maintain a central NDA repository and notify relevant employees of expiration dates.

Compliance & Penalties

Violations will result in mandatory retraining for a first offence; repeated or intentional breaches may lead to termination and legal action.

2.3 Policy 2 – Integrity: Data Modification and Version Control

Purpose:

To ensure the accuracy, consistency, and trustworthiness of data throughout its lifecycle.

Scope:

All personnel who create, edit, or delete company records, including developers and system administrators.

Policy Statements:

1. Unauthorised modification, alteration, or deletion of data is prohibited. All changes must be traceable.
2. Critical systems must enable audit logging, capturing the user ID, timestamp, and nature of every change.
3. File transfers (internal and external) must be verified using SHA-256 checksums to detect tampering.

4. All source code and official documents must be stored in version control systems (e.g., Git); merges require peer review.

Procedures:

- **Audit Log Review:** The IT security team will review logs bi-weekly and investigate anomalies.
- **Checksum Verification:** Automated scripts will validate checksums after every file transfer; failed transfers will be blocked.
- **Version Control Policy:** Developers must create feature branches and obtain at least one approval before merging into the main branch. The audit trail is preserved indefinitely.

Compliance & Penalties:

Any deliberate data tampering will be treated as gross misconduct, resulting in immediate termination and possible criminal referral.

2.4 Policy 3 – Availability: System Maintenance and Incident Response

Purpose:

To guarantee that information systems, applications, and data are accessible to authorised users when required.

Scope:

IT Operations, System Administrators, and all staff who depend on continuous service availability.

Policy Statements:

1. Critical databases must be backed up every 24 hours, with one copy stored in a geographically separate location.
2. Scheduled maintenance must be announced at least 48 hours in advance and carried out during off-peak hours (e.g., Saturday 02:00–06:00).

3. A Business Continuity Plan (BCP) must be activated within 30 minutes of a confirmed outage or denial-of-service attack, ensuring alternative workflows.

Procedures:

- **Backup Monitoring:** The backup system will send automated email confirmations; a monthly restore test will be performed.
- **Maintenance Windows:** A shared calendar (with automated reminders) will be maintained; changes are logged in a central change management system.
- **Incident Response:** The Incident Response Team (IRT) will follow the BCP checklist, post status updates on the internal portal, and conduct a post-mortem within 48 hours of resolution.

Compliance & Penalties:

Failure to follow maintenance protocols may result in disciplinary action. Negligence that causes prolonged downtime will be escalated to senior management.

2.5 Policy 4 – Authentication: Access Verification and Password Management

Purpose:

To verify the identity of any user attempting to access corporate networks, systems, or physical premises (OWASP Foundation, 2023).

Scope:

All employees, contractors, and temporary workers with any form of access credentials.

Policy Statements:

1. Multi-Factor Authentication (MFA) is mandatory for all remote access (VPN, email, cloud dashboards).
2. Passwords must be at least **14 characters**, including upper- and lowercase letters, digits, and special characters.

3. Passwords shall expire every **90 days**; the system must block reuse of the last **5 passwords**.
4. Sharing of credentials, tokens, or access badges is strictly forbidden.

Procedures:

- **MFA Enrolment:** HR will ensure MFA is configured during onboarding; IT will audit MFA usage quarterly.
- **Password Policy Enforcement:** Domain controllers will enforce complexity and history rules automatically.
- **Credential Hygiene:** Phishing simulations will be conducted monthly; users who fail will be assigned additional training.
- **Badge Management:** Lost badges must be reported within 2 hours; access will be immediately revoked.

Compliance & Penalties:

Sharing credentials constitutes a severe security violation. Offenders will face immediate suspension pending investigation and possible termination.

2.6 Policy 5 – Authorization: Least Privilege and Role-Based Access Control

Purpose:

To ensure users are granted only the minimum permissions necessary to perform their job functions.

Scope:

IT Administrators, HR, department managers, and all end-users with system access.

Policy Statements:

1. Access rights are assigned according to the **Principle of Least Privilege (PoLP)** and **Role-Based Access Control (RBAC)**.

2. Employees shall be granted access only to the folders, applications, and servers explicitly required by their role.
3. Any request for elevated privileges must be submitted via a formal ticket, approved by both the direct manager and the Information Security Officer.
4. The IT security team will conduct **semi-annual access reviews** to revoke outdated permissions (role changes, terminations).

Procedures:

- **RBAC Matrix:** HR and IT will maintain a matrix mapping job titles to resource permissions; updates occur within 3 business days of a role change.
- **Privilege Escalation Workflow:** Requests must include business justification and a defined expiry; temporary privileges auto-expire after 30 days unless renewed.
- **Access Review Audit:** A report will be generated every 6 months; discrepancies will be corrected within one week and documented.

Compliance & Penalties:

Unauthorised escalation of privileges is a critical violation and may lead to immediate dismissal.

2.7 Conclusion (Part 2)

These five policies establish a comprehensive governance framework that directly addresses the CIA triad and access control fundamentals. By combining clear policy statements with actionable procedures and defined consequences, the organisation minimises human-centric risks, promotes a culture of accountability, and aligns with industry best practices (National Institute of Standards and Technology [NIST], 2020). Regular review and employee awareness programmes will ensure that the policies remain effective and resilient against evolving cyber threats.

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MARKING RUBRIC: CRYPTOGRAPHIC SYSTEM

Category	Excellent (10-9)	Above Average (8-7)	Average (6-5)	Below Average (4-3)	Poor (2-1)	Total
System program features and UI design	<i>All of the menus, buttons, and links are fully-functional and attractive.</i>	<i>All of the menus, buttons, and links are fully-functional but less attractive.</i>	<i>Most of the menus, button, and links are fully-functional and attractive.</i>	<i>Most of the menus, button, and links are fully-functional but less attractive.</i>	<i>Most of the menus, buttons, and links are not fully-functional.</i>	/10
System complexity and code design	<i>Complex system and minimal code design, with or without comment.</i>	<i>Complex system but lengthy code design, with or without comment.</i>	<i>Simple system and minimal code design, with relevant comment added.</i>	<i>Simple system and minimal code design, but without comment.</i>	<i>Simple system but lengthy code design, with or without comment.</i>	/10
Encryption techniques	<i>Include more than 2 encryption techniques and accurately following the required methods.</i>	<i>Include more than 2 encryption techniques but inaccurately following the required methods.</i>	<i>Include 2 encryption techniques and accurately following the required methods.</i>	<i>Include 2 encryption techniques but inaccurately following the required methods.</i>	<i>Include less than 2 encryption techniques.</i>	/10
Encryption demonstration	<i>Well-documented demonstration of each encryption technique, with > 2 examples or samples.</i>	<i>Well-documented demonstration of each encryption technique, with 2 examples or samples.</i>	<i>Well-documented demonstration of each encryption technique, with 1 example or sample.</i>	<i>Minimally documented demonstration of each encryption technique, with > 1 example or sample.</i>	<i>Minimally documented demonstration of each encryption technique, with 1 example or sample.</i>	/10
Decryption demonstration	<i>Well-documented demonstration of each decryption technique, with > 2 examples or samples.</i>	<i>Well-documented demonstration of each decryption technique, with 2 examples or samples.</i>	<i>Well-documented demonstration of each decryption technique, with 1 example or sample.</i>	<i>Minimally documented demonstration of each decryption technique, with > 1 example or sample.</i>	<i>Minimally documented demonstration of each decryption technique, with 1 example or sample.</i>	/10
Report comprehension	<i>Report shows excellent understanding of the topic through a clear introduction and a meaningful conclusion.</i>	<i>Report shows good understanding of the topic with a clear introduction and an appropriate conclusion.</i>	<i>Report shows acceptable understanding of the topic with basic introduction and conclusion, but several sections were simple.</i>	<i>Report shows limited understanding of the topic with weak introduction and conclusion.</i>	<i>Report shows poor understanding of the topic with unclear, irrelevant, or incomplete introduction and conclusion.</i>	/10
Total						/60

MARKING RUBRIC: POLICIES AND PROCEDURES

Category	Excellent (10-9)	Above Average (8-7)	Average (6-5)	Below Average (4-3)	Poor (2-1)	Total
Policies and procedures design	All policy and procedure statements were relevant, clear, and well-designed.	Most of policy and procedure statements were relevant, clear, and well-designed.	All policy and procedure statements were relevant and adequately designed.	Most of policy and procedure statements were relevant and adequately designed.	Policy and procedure statements were unclear and lacked relevance.	/10
Policies and procedures scope	All policies and procedures statements were highly relevant, comprehensive, and well-designed.	Most policies and procedures statements were relevant and well-designed, although some were relatively simple.	Most policies and procedures statements were acceptable and relevant, but were generally simple with few well-developed statements.	Policies and procedures statements were generally acceptable but lacked depth, detail, and originality.	Most or some policies and procedures statements were irrelevant and poorly developed.	/10
Report comprehension	Report shows excellent understanding of the topic through a clear introduction and a meaningful conclusion.	Report shows good understanding of the topic with a clear introduction and an appropriate conclusion.	Report shows acceptable understanding of the topic with basic introduction and conclusion, but several sections were simple.	Report shows limited understanding of the topic with weak introduction and conclusion.	Report shows poor understanding of the topic with unclear, irrelevant, or incomplete introduction and conclusion.	/10
References citation	References and citations were complete, accurate, consistent, and properly formatted throughout the report.	Most references and citations were accurate and properly formatted, with only minor inconsistencies or errors.	References and citations were generally acceptable, with several formatting inconsistencies were present.	References and citations were limited, inconsistently formatted, or contained multiple errors.	References and citations were mostly missing, inaccurate, improperly formatted.	/10
Total						/40

OVERALL FEEDBACK

First Marker:

Total Mark: _____

Signature	Date

Second Marker:

Total Mark: _____

Signature	Date